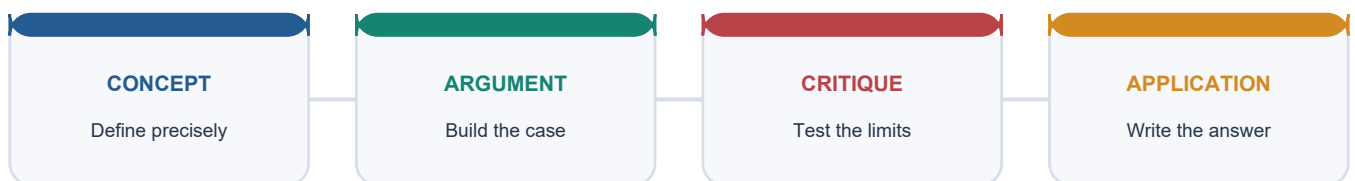


COMPLETE LEARNING SESSION

Algebra-Inequalities-and-Data-Sufficiency - CSAT Learner-v2 Semantic Successor

UPSC complete learning session | Foundation to advanced | Concepts, evidence, practice and answer writing

LEARNING PATH



Deterministic fallback visual: move from precise concepts through argument and critique to exam application.

Source order: repository knowledge -> official current linkage -> clearly marked analysis. No fabricated PYQs or statistics.

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Major learning parts and meaningful subtopics are listed in document order. Page numbers are generated from the final PDF layout; PDF bookmarks mirror the same hierarchy.

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Algebra-Inequalities-and-Data-Sufficiency - Complete CSAT Learning Session

Identity: csat-05: learner-v2:g1 **Generation date:** 2026-09-06 **Approval:** false **Official syllabus anchor:** Logical reasoning, analytical ability, decision making and Class X data sufficiency.

Source	SHA-256 at generation
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The canonical Basic owner is taught first. Optional Advanced material is isolated after practice. All official-PYQ references preserve the repository's supplied/provisional key labels; no unavailable wording, official explanation or official key is invented.

BASIC LEARNING SESSION

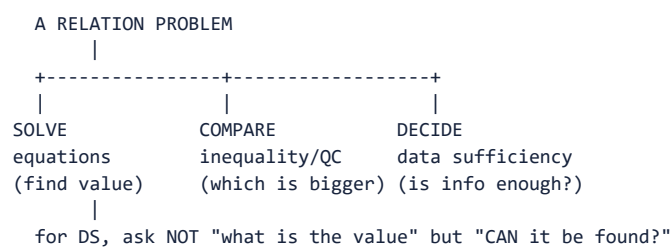
Twelve-panel ASCII master flow

PANEL 01 - EXPRESSION DISCIPLINE	
Track signs, brackets, domains and denominators.	
v	
PANEL 02 - LINEAR EQUATIONS	
Preserve equality through reversible operations.	
v	
PANEL 03 - SIMULTANEOUS EQUATIONS	
Use elimination, substitution or option testing.	
v	
PANEL 04 - INEQUALITIES	
Reverse the sign only when multiplying or dividing by a negative.	
v	
PANEL 05 - ABSOLUTE VALUE AND SURDS	
Split valid cases and respect non-negative roots.	
v	
PANEL 06 - WORD TRANSLATION	
Define variables and constraints before manipulating.	
v	
PANEL 07 - QUANTITATIVE COMPARISON	
Compare ranges, not one sample.	
v	
PANEL 08 - DS FORMAT	
Read the printed verdict options before solving.	
v	
PANEL 09 - NECESSARY AND SUFFICIENT	
Test each statement alone, then together.	
v	
PANEL 10 - COUNTEREXAMPLE SEARCH	
One second valid value disproves uniqueness.	
v	
PANEL 11 - BOUNDS AND OPTIMISATION	
Check endpoints, integrality and attainability.	
v	
PANEL 12 - VERIFICATION	
Substitute, test domains and separate answer value from sufficiency.	

Algebra, Inequalities and Data Sufficiency - MUST-DO

Subject: CSAT | **Tier:** Must-Do (foundation) | **Family:** Analytical ability / decision-making. **Core skill:** translate words to equations, handle inequalities and quantitative comparison, and master the UPSC two-statement **data-sufficiency** decision. **Grounded in:** audited UPSC CSAT PYQ sets 2024-2026 (Set A). Drills are **computationally verified**. **FACT:** = rule/fact | **ANALYSIS:** = guidance | □ = original example. *Companion: Advanced Drill.*

1. Visual foundation



Core proposition: data sufficiency tests a **decision** ("is the information enough?"), not a computation - the fastest solvers often **never** find the value.

2. The UPSC data-sufficiency format (read it, do not recall it)

FACT: A CSAT data-sufficiency item gives a **Question** followed by two **Statements I and II**. The first three options are stable across the audited papers; **the fourth is not**.

Option	Meaning (stable across 2024, 2025 and 2026 Set A)
(a)	Answerable using one statement alone, but not the other alone.
(b)	Answerable using either statement alone.
(c)	Answerable using both statements together , but not either alone.
(d)	ANALYSIS: Varies - read it . See the table below.

ANALYSIS: The fourth option is not a fixed template

Audited paper	Printed wording of (d)	What it means
2024 Set A (per item)	"The Question cannot be answered even by using both the Statements together"	Nothing works.
2025 Set A (some items)	"The Question cannot be answered even using any of the Statements"	Nothing works.
2025 Set A (other items)	"The Question can be answered even without using any of the Statements"	The question alone is self-sufficient .
2026 Set A (shared block of 5)	"the question cannot be answered even using any of the statements"	Nothing works.

- **FACT: Both 2025 variants appear in the same paper.** Two of the five audited 2025 data-sufficiency items print the "cannot" form and three print the "even without" form.
- **MEMORY: Consequence:** a candidate who marks (d) from memory as "nothing works" will mark the exact opposite verdict on a "can be answered without any statement" item. **Always read (d) before deciding.**
- **FACT: Layout also differs:** 2024 and 2025 repeat the directions and all four options ****inside every item**; **2026 prints them once**** above a block of five items. In 2026 you must scroll back to the block header to see what (d) says.

MEMORY: **DS discipline:** test **Statement I alone**, then **Statement II alone**, and only then **both together** - but before any of that, check whether the **question is already answerable on its own**, because in this paper that can be a live option. Do the value-finding **only** far enough to see whether the answer is **unique**.

Necessary vs sufficient (the distinction the format is built on)

Term	Test	In a DS item
FACT: Sufficient	Having it settles the question.	A statement is "sufficient" only if it forces exactly one answer.
FACT: Necessary	The conclusion cannot hold without it.	A statement can be necessary yet insufficient - it narrows without pinning.

- ANALYSIS: In $A \rightarrow B$: A is **sufficient** for B; B is **necessary** for A. They are not interchangeable, and a statement that is merely necessary is **not** enough to mark (a) or (b).

3. Essential algebra and inequality facts

Tool	Rule	Edge condition						
FACT: Linear solve	A linear equation in one unknown fixes it after isolation.	The variable coefficient must be non-zero; retain any stated integer/positive domain.						
FACT: Two unknowns	Two independent linear equations can pin two unknowns; a multiple of one is not independent.	Non-linear systems can still have 0, 1, or many solutions; inconsistent pairs give no solution and dependent pairs give many.						
FACT: Inequality flip	Multiplying/dividing by a negative reverses the sign.	Adding/subtracting never flips it.						
FACT: Dividing by a variable	Only legal if you know it is non-zero.	If its sign is unknown, the inequality direction is unknown too - split into cases; division by zero is never legal.						
FACT: Cross-multiplication	For $\frac{a}{b} < \frac{c}{d}$, multiply only after fixing b and d as non-zero and their signs.	With $bd > 0$, compare ad and bc; with $bd < 0$, reverse the inequality.						
FACT: Absolute value		u	$=k$ means $u=k$ or $u=-k$ when $k \geq 0$; for $k > 0$,	u	$<k$ means $-k < u < k$ and	u	$>k$ means $u < -k$ or $u > k$.	If $k < 0$, no real solution; handle $k=0$ separately.
FACT: Squaring to compare	$a > b \geq 0 \Rightarrow a^2 > b^2$.	Fails if either side may be negative.						
FACT: Quantitative comparison	Reduce both quantities to the same base/form before comparing.	x^m vs x^n reverses for $0 < x < 1$.						

Tool	Rule	Edge condition						
FACT: Sum-difference	Two numbers with sum S, difference D: they are $(S+D)/2$ and $(S-D)/2$.	Integer answers need S and D of the same parity.						

4. Method

1. **Solve items:** translate each sentence to one equation; write the domain first (integer, digit, positive, non-zero); then count unknowns vs independent **linear** equations.
2. **Inequality/QC items:** rewrite both sides in a comparable form (same base, same power); watch the sign rule and any denominator.
3. **DS items - in this order:**
 1. **Read the four printed options**, especially (d).
 2. Ask whether the **question alone** already pins the answer (a live verdict in this paper).
 3. Test **I alone**, then **II alone**, then **both** - stopping at **uniqueness**, not the value.

5. Original solved examples

□ Example A (solve)

Solve $3x + 5 = 20$. $3x = 15 \rightarrow x = 5$. (Verified.)

□ Example B (sum-difference)

Two numbers have sum 30 and difference 8. They are $(30+8)/2 = 19$ and $(30-8)/2 = 11$. **19 and 11**. (Verified.)

□ Example C (quantitative comparison)

Which is larger, 2^{30} or 3^{20} ? $2^{30} = 8^{10}$, $3^{20} = 9^{10} \rightarrow 9^{10} > 8^{10} \rightarrow 3^{20}$ is larger. (Verified: $1073741824 < 3486784401$.)

□ Example C.1 (word translation, denominator and absolute value)

A number increased by 5 is three times the number decreased by 1. Find it. Translate the words before calculating: $x + 5 = 3(x - 1)$, so $x + 5 = 3x - 3$, $2x = 8$, and $x = 4$.

Solve $|2x - 1| = 5$. Keep both branches: $2x - 1 = 5$ gives $x = 3$; $2x - 1 = -5$ gives $x = -2$. **Both values work**. (Verified.)

Compare $1/x$ and $1/y$ when $x > y > 0$. Because $xy > 0$, cross-multiplication gives $y < x$, hence $1/x < 1/y$. This conclusion is unsafe without the positive-domain condition.

□ Example D (data sufficiency, answer (c))

Question: What is the two-digit number N? Statement I: N is divisible by 11. Statement II: The tens digit of N is 3.
I alone $\rightarrow$ 11, 22, ..., 99 (not unique). II alone $\rightarrow$ 30-39 (not unique). Both $\rightarrow$ divisible by 11 **and** tens digit 3 $\rightarrow$ **33** (unique). **Answer: (c)**.

□ Example E (data sufficiency, answer (b))

Question: Is the integer n even? Statement I: n^2 is even. Statement II: $n + 1$ is odd.

I alone $\rightarrow$ n^2 even forces n even (sufficient). II alone $\rightarrow$ $n+1$ odd forces n even (sufficient). Each works alone. **Answer: (b)**.

□ Example F (the self-sufficient question - the 2025 fourth-option variant)

Question: What is the smallest **two-digit** number having exactly **three** distinct factors? Statement I: the number is odd. Statement II: the number is divisible by 5.

Why the factor shortcut is safe here: if $n = p^a q^b \dots$, its number of positive factors is $(a+1)(b+1)\dots$. Exactly three factors therefore forces one exponent 2 and no other prime: $n = p^2$. A number has exactly three distinct factors only if it is the **square of a prime** ($1, p, p^2$). Two-digit prime squares are 25 and 49, so the

smallest is **25** - and that came from the **question alone**. Statement I is true of 25 and Statement II is true of 25, but **neither was needed**.

Verdict: the question is answerable **without** either statement. Under the 2025-style fourth option ("The Question can be answered even **without** using any of the Statements") that is **(d)**. Under a 2024-style fourth option ("cannot be answered even by using both together") the same reasoning would make (d) **wrong** and the printed set would have to be re-read. (Verified: divisors of 25 = {1, 5, 25}; of 49 = {1, 7, 49}; no smaller two-digit number has exactly three.)

MEMORY: This is the whole lesson of Section 2: the same reasoning maps to different letters depending on what is printed. Solve the logic, then read the label.

6. Must-Know facts

- FACT: Two independent linear equations are needed to pin two unknowns; $2x + 4y = 20$ adds **nothing** to $x + 2y = 10$.

- FACT: In DS, a statement is "sufficient" only if it yields a **single** answer; a statement that merely narrows the field is **necessary at best**, never sufficient.

- FACT: **Sufficiency is about the answer, not about the arithmetic**. Never finish the calculation.

- FACT: Reversing an inequality happens on multiply/divide by a negative - **not** on adding a negative.

Adding the same quantity to both sides never flips the sign.

- FACT: Squaring both sides preserves order **only when both sides are non-negative**.

- FACT: For QC, common tricks: equalise exponents, take roots, or compare ratios.

- FACT: Before cross-multiplying an inequality, state whether the denominator product is positive or negative; zero denominators are excluded.

- FACT: A number that is both a perfect square and cube has every prime exponent divisible by both 2 and 3, hence by 6; this supports the self-sufficiency drill without a hidden prerequisite.

7. Common traps

- WRONG: Computing the value in a DS item when you only need "is it unique?". -> Stop at uniqueness.

- WRONG: **Marking (d) from memory**. -> The printed (d) differs between the audited years and even between items of the same 2025 paper; read it.

- WRONG: Missing that the **question alone** may already be answerable. -> Check that first.

- WRONG: Treating a scaled copy of an equation as new information. -> It is dependent.

- WRONG: Confusing **necessary** with **sufficient** ("n is even" is necessary for "n is a multiple of 6", but nowhere near sufficient). -> Ask which direction the arrow points.

- WRONG: Forgetting to flip the inequality on a negative multiplier. -> Flip the sign.

- WRONG: Cross-multiplying before checking a denominator sign. -> State $bd > 0$ or split cases.

- WRONG: Keeping only one branch of an absolute-value equation. -> Write $u = k$ and $u = -k$.

- WRONG: In DS, checking "both together" **before** testing each alone. -> Test each alone first (it may be (a) or (b)).

- WRONG: Assuming "cannot be found" without trying **both together**. -> That verdict requires both to fail.

8. Quick checks

- FACT: Can you state options (a), (b) and (c) - and did you just **read** (d) off the paper?
- FACT: Do you test I alone and II alone **before** combining?
- FACT: Can you say which of two conditions is necessary and which is sufficient?
- FACT: Can you compare a^m and b^n by equalising exponents?

9. Mini-drill (with answers and explanations)

1. Solve the inequality $2x - 3 > 7$.
2. Compare $(0.5)^3$ and $(0.5)^2$.
3. **DS.** Question: What is the two-digit number? I: The sum of its digits is 9. II: The difference of its digits is 9.
4. **DS.** Question: What is x ? I: $3x = 12$. II: $x + 7 = 11$.
5. Two numbers have sum 30 and difference 8. Find them.
6. For the claim "n is a multiple of 12", is "n is even" **necessary**, **sufficient**, both, or neither?
7. **DS.** Question: What is the smallest integer greater than 1 that is both a perfect square and a perfect cube? I: it is even. II: it is less than 100.

Answers.

1. **$x > 5$.** Add 3, divide by 2. (*Verified.*)
2. $(0.5)^3 < (0.5)^2$ ($0.125 < 0.25$): for a base between 0 and 1, a higher power is smaller. (*Verified.*)
3. **(a).** I alone: 18, 27, 36, ..., 90 (many). II alone: digits differing by 9 -> only **90** (since 9 and 0), unique. So one statement (II) alone answers it but not the other. (*Verified: only 90.*)
4. **(b).** I alone -> $x = 4$; II alone -> $x = 4$. Either works. (*Verified.*)
5. **19 and 11.** $(30 \pm 8) / 2$. (*Verified.*)
6. **Necessary but not sufficient.** Every multiple of 12 is even, so evenness cannot be dropped; but 14 is even and not a multiple of 12, so evenness does not settle it. (*Verified counter-example.*)
7. **The question is self-sufficient.** A number that is both a square and a cube is a **6th power**; the smallest above 1 is $2^6 = 64$. Neither statement was used - both merely happen to be true of 64. Under the 2025-style fourth option that is **(d)**; under a 2024-style fourth option it is not, so **read the printed options** before marking. (*Verified: $64 = 8^2 = 4^3$; next is $3^6 = 729$.*)

10. Timed transfer, diagnosis and retry gate

Set 90 seconds per item; do not look at a DS letter until the logical verdict is fixed.

1. Solve $3(x - 2) = 2x + 7$ for integer x .
2. Solve $|x + 4| = 2$.
3. Given $a > b > 0$, compare a/b and b/a .
4. **DS.** Is positive integer n divisible by 6? I: n is even. II: n is divisible by 3.
5. A two-digit number is $10t + u$. Its digits sum to 11 and its tens digit exceeds its units digit by 3. Find the number.

Answers. 1. **13.** 2. **-2 or -6.** 3. $a/b > b/a$ because multiplying by $ab > 0$ reduces it to $a^2 > b^2$. 4. **Both needed.** 5. $t+u=11$, $t-u=3$, so $t=7$, $u=4$: **74.**

If you missed because...	Code	Repair before retry
You assumed two equations always give one answer	C	State "independent linear equations + domain".
You lost a sign/denominator restriction	A	Write $x \neq 0$ and denominator signs first.
You kept one absolute-value branch	C	Draw the two branches before solving.
You mapped a DS verdict to a memorised letter	R	Re-read the printed option block.
You over-solved a DS item	T	Stop once uniqueness or a counterexample is proved.

Retry gate: redo the five items after 20 minutes. Continue only at **5/5 within eight minutes**; otherwise rebuild the failed relation and take five fresh items of that exact form.

11. Study links

- FACT: Optional Advanced companion -

dependent-statement DS, unique-root logic, and interval/optimisation depth. Core above remains sufficient for the paper's basic algebra and DS forms.

- FACT: Reading Comprehension - "is it supported?" mirrors "is it sufficient?".
- FACT: Number Systems and Number Sense - number properties power many DS items.

2026 PYQ Integration

Status: 2026 question-level PYQ demand is integrated into this owner. **Provenance:** Audited local official-paper routing ledgers: `_PYQ-ROUTING-CSAT-2026.md`. **Answer-key rule:** The 2026 Prelims and CSAT Set-A keys held locally are **provisional**; no option or answer is recorded or inferred in this integration.

- **Year represented:** 2026
- **Paper(s):** CSAT
- **Routed question demands:** 9

Year	Paper	Q	PYQ demand (neutral rendering)	Directive / format	Source status	Owner requirement
2026	CSAT	1	Data sufficiency: coin denomination	Objective question; provisional 2026 Set-A key present locally, answer not inferred	Provisional 2026 Set-A key present locally (scan headed CS (P) Exam 2026 [Prov. Ans. Key]); family/neutral type per CSAT/00_Question-Audit-Ledger; key is provisional - no answer letter recorded or inferred here	Practise this exact skill form under timed elimination; no answer is inferred here.
2026	CSAT	2	Data sufficiency: compare reals	Objective question; provisional 2026 Set-A key present locally, answer not inferred	Provisional 2026 Set-A key present locally (scan headed CS (P) Exam 2026 [Prov. Ans. Key]); family/neutral type per CSAT/00_Question-Audit-Ledger; key is provisional - no answer letter recorded or inferred here	Practise this exact skill form under timed elimination; no answer is inferred here.

Year	Paper	Q	PYQ demand (neutral rendering)	Directive / format	Source status	Owner requirement
2026	CSAT	3	Data sufficiency: parity	Objective question; provisional 2026 Set-A key present locally, answer not inferred	Provisional 2026 Set-A key present locally (scan headed CS (P) Exam 2026 [Prov. Ans. Key]); family/neutral type per CSAT/00_Question-Audit-Ledger; key is provisional - no answer letter recorded or inferred here	Practise this exact skill form under timed elimination; no answer is inferred here.
2026	CSAT	4	Data sufficiency: set-membership logic	Objective question; provisional 2026 Set-A key present locally, answer not inferred	Provisional 2026 Set-A key present locally (scan headed CS (P) Exam 2026 [Prov. Ans. Key]); family/neutral type per CSAT/00_Question-Audit-Ledger; key is provisional - no answer letter recorded or inferred here	Practise this exact skill form under timed elimination; no answer is inferred here.
2026	CSAT	5	Data sufficiency: prime determination	Objective question; provisional 2026 Set-A key present locally, answer not inferred	Provisional 2026 Set-A key present locally (scan headed CS (P) Exam 2026 [Prov. Ans. Key]); family/neutral type per CSAT/00_Question-Audit-Ledger; key is provisional - no answer letter recorded or inferred here	Practise this exact skill form under timed elimination; no answer is inferred here.
2026	CSAT	8	Weight equations comparison	Objective question; provisional 2026 Set-A key present locally, answer not inferred	Provisional 2026 Set-A key present locally (scan headed CS (P) Exam 2026 [Prov. Ans. Key]); family/neutral type per CSAT/00_Question-Audit-Ledger; key is provisional - no answer letter recorded or inferred here	Practise this exact skill form under timed elimination; no answer is inferred here.
2026	CSAT	21	Inequality-range statements	Objective question; provisional 2026 Set-A key present locally, answer not inferred	Provisional 2026 Set-A key present locally (scan headed CS (P) Exam 2026 [Prov. Ans. Key]); family/neutral type per CSAT/00_Question-Audit-Ledger; key is provisional - no answer letter recorded or inferred here	Practise this exact skill form under timed elimination; no answer is inferred here.
2026	CSAT	38	Linear marking-scheme equations	Objective question; provisional 2026 Set-A key present locally, answer not inferred	Provisional 2026 Set-A key present locally (scan headed CS (P) Exam 2026 [Prov. Ans. Key]); family/neutral type per CSAT/00_Question-Audit-Ledger; key is provisional - no answer letter recorded or inferred here	Practise this exact skill form under timed elimination; no answer is inferred here.

Year	Paper	Q	PYQ demand (neutral rendering)	Directive / format	Source status	Owner requirement
2026	CSAT	80	Rate-constraint optimisation	Objective question; provisional 2026 Set-A key present locally, answer not inferred	Provisional 2026 Set-A key present locally (scan headed CS (P) Exam 2026 [Prov. Ans. Key]); family/neutral type per CSAT/00_Question-Audit-Ledger; key is provisional - no answer letter recorded or inferred here	Practise this exact skill form under timed elimination; no answer is inferred here.

What this owner must now support

- Data sufficiency: coin denomination
- Data sufficiency: compare reals
- Data sufficiency: parity
- Data sufficiency: set-membership logic
- Data sufficiency: prime determination
- Weight equations comparison
- Inequality-range statements
- Linear marking-scheme equations
- Rate-constraint optimisation

This block integrates the 2026 examinable demand and paper metadata. It is kept separate from the 2018-2023 and 2024-2025 blocks and does not convert a provisionally-keyed, answer-free objective question into a solved answer.

Recent PYQ Integration (2024-2025)

Status: 2024-2025 question-level PYQ demand is integrated into this owner. **Provenance:** Audited local official-paper routing ledgers: _PYQ-ROUTING-CSAT-2024-2025.md. **Answer-key rule:** The official 2024-2025 Prelims Set-A keys are present in the repository and CSAT Set-A keys are supplied; even so, no option or answer is recorded or inferred in this integration.

- **Years represented:** 2024, 2025
- **Paper(s):** CSAT
- **Routed question demands:** 24

Year	Paper	Q	PYQ demand (neutral rendering)	Directive / format	Source status	Owner requirement
2024	CSAT	35	Reversed-digit equation	Objective question; official Set-A key available locally, answer not inferred	Key supplied locally (Set-A scan; recorded as supplied, not certified final); family and neutral type per CSAT/00_Question-Audit-Ledger; answer not recorded here	Practise this exact skill form under timed elimination; no answer is inferred here.
2024	CSAT	50	Linear-cost validation	Objective question; official Set-A key available locally, answer not inferred	Key supplied locally (Set-A scan; recorded as supplied, not certified final); family and neutral type per CSAT/00_Question-Audit-Ledger; answer not recorded here	Practise this exact skill form under timed elimination; no answer is inferred here.

Year	Paper	Q	PYQ demand (neutral rendering)	Directive / format	Source status	Owner requirement
2024	CSAT	59	Data sufficiency: integers	Objective question; official Set-A key available locally, answer not inferred	Key supplied locally (Set-A scan; recorded as supplied, not certified final); family and neutral type per CSAT/00_Question-Audit-Ledger; answer not recorded here	Practise this exact skill form under timed elimination; no answer is inferred here.
2024	CSAT	60	Data sufficiency: transfer rate	Objective question; official Set-A key available locally, answer not inferred	Key supplied locally (Set-A scan; recorded as supplied, not certified final); family and neutral type per CSAT/00_Question-Audit-Ledger; answer not recorded here	Practise this exact skill form under timed elimination; no answer is inferred here.
2024	CSAT	64	Data sufficiency: integer pair	Objective question; official Set-A key available locally, answer not inferred	Key supplied locally (Set-A scan; recorded as supplied, not certified final); family and neutral type per CSAT/00_Question-Audit-Ledger; answer not recorded here	Practise this exact skill form under timed elimination; no answer is inferred here.
2024	CSAT	65	Data sufficiency: shares	Objective question; official Set-A key available locally, answer not inferred	Key supplied locally (Set-A scan; recorded as supplied, not certified final); family and neutral type per CSAT/00_Question-Audit-Ledger; answer not recorded here	Practise this exact skill form under timed elimination; no answer is inferred here.
2024	CSAT	66	Data sufficiency: class size	Objective question; official Set-A key available locally, answer not inferred	Key supplied locally (Set-A scan; recorded as supplied, not certified final); family and neutral type per CSAT/00_Question-Audit-Ledger; answer not recorded here	Practise this exact skill form under timed elimination; no answer is inferred here.
2024	CSAT	67	Data sufficiency: prime triple	Objective question; official Set-A key available locally, answer not inferred	Key supplied locally (Set-A scan; recorded as supplied, not certified final); family and neutral type per CSAT/00_Question-Audit-Ledger; answer not recorded here	Practise this exact skill form under timed elimination; no answer is inferred here.
2024	CSAT	68	Data sufficiency: integrality	Objective question; official Set-A key available locally, answer not inferred	Key supplied locally (Set-A scan; recorded as supplied, not certified final); family and neutral type per CSAT/00_Question-Audit-Ledger; answer not recorded here	Practise this exact skill form under timed elimination; no answer is inferred here.
2024	CSAT	69	Data sufficiency: article prices	Objective question; official Set-A key available locally, answer not inferred	Key supplied locally (Set-A scan; recorded as supplied, not certified final); family and neutral type per CSAT/00_Question-Audit-Ledger; answer not recorded here	Practise this exact skill form under timed elimination; no answer is inferred here.

Year	Paper	Q	PYQ demand (neutral rendering)	Directive / format	Source status	Owner requirement
2024	CSAT	70	Data sufficiency: score comparison	Objective question; official Set-A key available locally, answer not inferred	Key supplied locally (Set-A scan; recorded as supplied, not certified final); family and neutral type per CSAT/00_Question-Audit-Ledger; answer not recorded here	Practise this exact skill form under timed elimination; no answer is inferred here.
2024	CSAT	73	Data sufficiency: reversed ages	Objective question; official Set-A key available locally, answer not inferred	Key supplied locally (Set-A scan; recorded as supplied, not certified final); family and neutral type per CSAT/00_Question-Audit-Ledger; answer not recorded here	Practise this exact skill form under timed elimination; no answer is inferred here.
2024	CSAT	76	Coded inequalities	Objective question; official Set-A key available locally, answer not inferred	Key supplied locally (Set-A scan; recorded as supplied, not certified final); family and neutral type per CSAT/00_Question-Audit-Ledger; answer not recorded here	Practise this exact skill form under timed elimination; no answer is inferred here.
2025	CSAT	19	Chained-inequality validation	Objective question; official Set-A key available locally, answer not inferred	Key supplied locally (Set-A scan; recorded as supplied, not certified final); family and neutral type per CSAT/00_Question-Audit-Ledger; answer not recorded here	Practise this exact skill form under timed elimination; no answer is inferred here.
2025	CSAT	26	Bounded-expression ratio	Objective question; official Set-A key available locally, answer not inferred	Key supplied locally (Set-A scan; recorded as supplied, not certified final); family and neutral type per CSAT/00_Question-Audit-Ledger; answer not recorded here	Practise this exact skill form under timed elimination; no answer is inferred here.
2025	CSAT	68	Data sufficiency: factor count	Objective question; official Set-A key available locally, answer not inferred	Key supplied locally (Set-A scan; recorded as supplied, not certified final); family and neutral type per CSAT/00_Question-Audit-Ledger; answer not recorded here	Practise this exact skill form under timed elimination; no answer is inferred here.
2025	CSAT	69	Data sufficiency: digit product	Objective question; official Set-A key available locally, answer not inferred	Key supplied locally (Set-A scan; recorded as supplied, not certified final); family and neutral type per CSAT/00_Question-Audit-Ledger; answer not recorded here	Practise this exact skill form under timed elimination; no answer is inferred here.
2025	CSAT	70	Data sufficiency: family relation	Objective question; official Set-A key available locally, answer not inferred	Key supplied locally (Set-A scan; recorded as supplied, not certified final); family and neutral type per CSAT/00_Question-Audit-Ledger; answer not recorded here	Practise this exact skill form under timed elimination; no answer is inferred here.

Year	Paper	Q	PYQ demand (neutral rendering)	Directive / format	Source status	Owner requirement
2025	CSAT	73	Data sufficiency: match outcome	Objective question; official Set-A key available locally, answer not inferred	Key supplied locally (Set-A scan; recorded as supplied, not certified final); family and neutral type per CSAT/00_Question-Audit-Ledger; answer not recorded here	Practise this exact skill form under timed elimination; no answer is inferred here.
2025	CSAT	74	Data sufficiency: positivity	Objective question; official Set-A key available locally, answer not inferred	Key supplied locally (Set-A scan; recorded as supplied, not certified final); family and neutral type per CSAT/00_Question-Audit-Ledger; answer not recorded here	Practise this exact skill form under timed elimination; no answer is inferred here.
2025	CSAT	75	Quantitative comparison: ratio	Objective question; official Set-A key available locally, answer not inferred	Key supplied locally (Set-A scan; recorded as supplied, not certified final); family and neutral type per CSAT/00_Question-Audit-Ledger; answer not recorded here	Practise this exact skill form under timed elimination; no answer is inferred here.
2025	CSAT	76	Quantitative comparison: product	Objective question; official Set-A key available locally, answer not inferred	Key supplied locally (Set-A scan; recorded as supplied, not certified final); family and neutral type per CSAT/00_Question-Audit-Ledger; answer not recorded here	Practise this exact skill form under timed elimination; no answer is inferred here.
2025	CSAT	77	Quantitative comparison: set	Objective question; official Set-A key available locally, answer not inferred	Key supplied locally (Set-A scan; recorded as supplied, not certified final); family and neutral type per CSAT/00_Question-Audit-Ledger; answer not recorded here	Practise this exact skill form under timed elimination; no answer is inferred here.
2025	CSAT	79	Unit-interval inequalities	Objective question; official Set-A key available locally, answer not inferred	Key supplied locally (Set-A scan; recorded as supplied, not certified final); family and neutral type per CSAT/00_Question-Audit-Ledger; answer not recorded here	Practise this exact skill form under timed elimination; no answer is inferred here.

What this owner must now support

- Reversed-digit equation
- Linear-cost validation
- Data sufficiency: integers
- Data sufficiency: transfer rate
- Data sufficiency: integer pair
- Data sufficiency: shares
- Data sufficiency: class size
- Data sufficiency: prime triple
- Data sufficiency: integrality
- Data sufficiency: article prices
- Data sufficiency: score comparison
- Data sufficiency: reversed ages

- Coded inequalities
- Chained-inequality validation
- Bounded-expression ratio
- Data sufficiency: factor count
- Data sufficiency: digit product
- Data sufficiency: family relation
- Data sufficiency: match outcome
- Data sufficiency: positivity
- Quantitative comparison: ratio
- Quantitative comparison: product
- Quantitative comparison: set
- Unit-interval inequalities

This block integrates the 2024-2025 examinable demand and paper metadata. It is kept separate from the 2018-2023 block and does not convert an unkeyed/answer-free objective question into a solved answer.

Historical PYQ Integration (2018-2023)

Status: Question-level PYQ demand is integrated into this owner. **Provenance:** Audited local official-paper routing ledgers: `_PYQ-ROUTING-CSAT-2018-2023.md`. **Answer-key rule:** The official 2018-2023 Prelims/CSAT keys are not held locally; no option or answer has been inferred.

- **Years represented:** 2018, 2019, 2020, 2021, 2022, 2023
- **Paper(s):** CSAT
- **Routed question demands:** 41

Year	Paper	Q	PYQ demand (neutral rendering)	Directive / format	Source status	Owner requirement
2018	CSAT	2	Series equation solving	Objective question; official key unavailable locally	Stem verified against official scan; OCR artifact resolved; key unavailable locally	Practise this exact skill form under timed elimination; no answer is inferred here.
2018	CSAT	23	Range inequality deduction	Objective question; official key unavailable locally	Routed; key unavailable locally	Practise this exact skill form under timed elimination; no answer is inferred here.
2018	CSAT	24	Expression magnitude comparison	Objective question; official key unavailable locally	Routed; key unavailable locally	Practise this exact skill form under timed elimination; no answer is inferred here.
2018	CSAT	34	Digit-constraint algebra	Objective question; official key unavailable locally	Routed; key unavailable locally	Practise this exact skill form under timed elimination; no answer is inferred here.
2018	CSAT	37	Inequality chain deduction	Objective question; official key unavailable locally	Routed; key unavailable locally	Practise this exact skill form under timed elimination; no answer is inferred here.
2018	CSAT	38	Algebraic revenue expression	Objective question; official key unavailable locally	Stem verified against official scan; OCR artifact resolved; key unavailable locally	Practise this exact skill form under timed elimination; no answer is inferred here.

Year	Paper	Q	PYQ demand (neutral rendering)	Directive / format	Source status	Owner requirement
2018	CSAT	77	Equation implication deduction	Objective question; official key unavailable locally	Routed; key unavailable locally	Practise this exact skill form under timed elimination; no answer is inferred here.
2019	CSAT	18	Age ratio equations	Objective question; official key unavailable locally	Routed; key unavailable locally	Practise this exact skill form under timed elimination; no answer is inferred here.
2019	CSAT	37	Bounded inequality range	Objective question; official key unavailable locally	Routed; key unavailable locally	Practise this exact skill form under timed elimination; no answer is inferred here.
2019	CSAT	38	Age variable algebra	Objective question; official key unavailable locally	Routed; key unavailable locally	Practise this exact skill form under timed elimination; no answer is inferred here.
2019	CSAT	50	Fraction comparison property	Objective question; official key unavailable locally	Routed; key unavailable locally	Practise this exact skill form under timed elimination; no answer is inferred here.
2019	CSAT	58	Two-statement sufficiency	Objective question; official key unavailable locally	Routed; key unavailable locally	Practise this exact skill form under timed elimination; no answer is inferred here.
2019	CSAT	69	Piece-length ratio algebra	Objective question; official key unavailable locally	Routed; key unavailable locally	Practise this exact skill form under timed elimination; no answer is inferred here.
2020	CSAT	9	Equal-expression ordering	Objective question; official key unavailable locally	Routed; key unavailable locally	Practise this exact skill form under timed elimination; no answer is inferred here.
2020	CSAT	16	Digit-constraint sum puzzle	Objective question; official key unavailable locally	Routed; key unavailable locally	Practise this exact skill form under timed elimination; no answer is inferred here.
2020	CSAT	17	Two-statement sufficiency	Objective question; official key unavailable locally	Routed; key unavailable locally	Practise this exact skill form under timed elimination; no answer is inferred here.
2020	CSAT	28	Two-statement sufficiency	Objective question; official key unavailable locally	Routed; key unavailable locally	Practise this exact skill form under timed elimination; no answer is inferred here.
2020	CSAT	29	Two-statement sufficiency	Objective question; official key unavailable locally	Routed; key unavailable locally	Practise this exact skill form under timed elimination; no answer is inferred here.
2020	CSAT	30	Two-statement sufficiency	Objective question; official key unavailable locally	Routed; key unavailable locally	Practise this exact skill form under timed elimination; no answer is inferred here.
2020	CSAT	79	Vessel weight linear equation	Objective question; official key unavailable locally	Routed; key unavailable locally	Practise this exact skill form under timed elimination; no answer is inferred here.

Year	Paper	Q	PYQ demand (neutral rendering)	Directive / format	Source status	Owner requirement
2021	CSAT	8	Exponential model identification	Objective question; official key unavailable locally	Routed; key unavailable locally	Practise this exact skill form under timed elimination; no answer is inferred here.
2021	CSAT	38	Two-statement sufficiency	Objective question; official key unavailable locally	Routed; key unavailable locally	Practise this exact skill form under timed elimination; no answer is inferred here.
2021	CSAT	39	Two-statement sufficiency	Objective question; official key unavailable locally	Routed; key unavailable locally	Practise this exact skill form under timed elimination; no answer is inferred here.
2021	CSAT	47	Two-statement sufficiency	Objective question; official key unavailable locally	Routed; key unavailable locally	Practise this exact skill form under timed elimination; no answer is inferred here.
2021	CSAT	58	Two-variable score algebra	Objective question; official key unavailable locally	Routed; key unavailable locally	Practise this exact skill form under timed elimination; no answer is inferred here.
2021	CSAT	77	Age-relation algebra	Objective question; official key unavailable locally	Routed; key unavailable locally	Practise this exact skill form under timed elimination; no answer is inferred here.
2022	CSAT	19	Two-statement sufficiency	Objective question; official key unavailable locally	Routed; key unavailable locally	Practise this exact skill form under timed elimination; no answer is inferred here.
2022	CSAT	25	Two-statement sufficiency	Objective question; official key unavailable locally	Routed; key unavailable locally	Practise this exact skill form under timed elimination; no answer is inferred here.
2022	CSAT	28	Reverse chain equation	Objective question; official key unavailable locally	Routed; key unavailable locally	Practise this exact skill form under timed elimination; no answer is inferred here.
2022	CSAT	37	Two-statement sufficiency	Objective question; official key unavailable locally	Routed; key unavailable locally	Practise this exact skill form under timed elimination; no answer is inferred here.
2022	CSAT	48	Two-statement sufficiency	Objective question; official key unavailable locally	Routed; key unavailable locally	Practise this exact skill form under timed elimination; no answer is inferred here.
2022	CSAT	49	Two-statement sufficiency	Objective question; official key unavailable locally	Routed; key unavailable locally	Practise this exact skill form under timed elimination; no answer is inferred here.
2022	CSAT	50	Two-statement sufficiency	Objective question; official key unavailable locally	Routed; key unavailable locally	Practise this exact skill form under timed elimination; no answer is inferred here.
2022	CSAT	60	Percentage inequality comparison	Objective question; official key unavailable locally	Routed; key unavailable locally	Practise this exact skill form under timed elimination; no answer is inferred here.

Year	Paper	Q	PYQ demand (neutral rendering)	Directive / format	Source status	Owner requirement
2022	CSAT	78	Consecutive integer equation	Objective question; official key unavailable locally	Routed; key unavailable locally	Practise this exact skill form under timed elimination; no answer is inferred here.
2023	CSAT	26	Two-group sum product maximum	Objective question; official key unavailable locally	Routed; key unavailable locally	Practise this exact skill form under timed elimination; no answer is inferred here.
2023	CSAT	56	Two-statement inequality sufficiency	Objective question; official key unavailable locally	Routed; key unavailable locally	Practise this exact skill form under timed elimination; no answer is inferred here.
2023	CSAT	57	Two-statement integer sufficiency	Objective question; official key unavailable locally	Routed; key unavailable locally	Practise this exact skill form under timed elimination; no answer is inferred here.
2023	CSAT	58	Two-statement set-size sufficiency	Objective question; official key unavailable locally	Routed; key unavailable locally	Practise this exact skill form under timed elimination; no answer is inferred here.
2023	CSAT	59	Two-statement digit-number sufficiency	Objective question; official key unavailable locally	Stem verified against official scan; OCR artifact resolved; key unavailable locally	Practise this exact skill form under timed elimination; no answer is inferred here.
2023	CSAT	60	Two-statement age-sequence sufficiency	Objective question; official key unavailable locally	Stem verified against official scan; OCR artifact resolved; key unavailable locally	Practise this exact skill form under timed elimination; no answer is inferred here.

What this owner must now support

- Series equation solving
- Range inequality deduction
- Expression magnitude comparison
- Digit-constraint algebra
- Inequality chain deduction
- Algebraic revenue expression
- Equation implication deduction
- Age ratio equations
- Bounded inequality range
- Age variable algebra
- Fraction comparison property
- Two-statement sufficiency
- Piece-length ratio algebra
- Equal-expression ordering
- Digit-constraint sum puzzle
- Vessel weight linear equation
- Exponential model identification
- Two-variable score algebra
- Age-relation algebra
- Reverse chain equation

- Percentage inequality comparison
- Consecutive integer equation
- Two-group sum product maximum
- Two-statement inequality sufficiency
- Two-statement integer sufficiency
- Two-statement set-size sufficiency
- Two-statement digit-number sufficiency
- Two-statement age-sequence sufficiency

The table integrates the examinable demand and paper metadata. It does not turn an unkeyed objective question into a solved answer, and it does not claim that lexical presence alone proves full conceptual sufficiency.

Semantic-completeness closure - 2026-09-06

Literal syllabus and canonical ownership

- **Literal clause:** Logical reasoning, analytical ability, decision making and Class X data sufficiency.
- **Canonical scope:** Owns algebraic expressions, equations, inequalities, quantitative comparison and the two-statement data-sufficiency decision format.
- **Cross-topic boundary:** Arithmetic computation remains with Topic 03 and rate contexts with Topic 04 when no sufficiency judgement is tested; arrangements and coding belong to Topic 06.

Complete learner route

1. **Expression discipline:** Track signs, brackets, domains and denominators.
2. **Linear equations:** Preserve equality through reversible operations.
3. **Simultaneous equations:** Use elimination, substitution or option testing.
4. **Inequalities:** Reverse the sign only when multiplying or dividing by a negative.
5. **Absolute value and surds:** Split valid cases and respect non-negative roots.
6. **Word translation:** Define variables and constraints before manipulating.
7. **Quantitative comparison:** Compare ranges, not one sample.
8. **DS format:** Read the printed verdict options before solving.
9. **Necessary and sufficient:** Test each statement alone, then together.
10. **Counterexample search:** One second valid value disproves uniqueness.
11. **Bounds and optimisation:** Check endpoints, integrality and attainability.
12. **Verification:** Substitute, test domains and separate answer value from sufficiency.

Verification and hostile-query gate

Equations are plug-checked, inequality sign changes are explicit, and sufficiency verdicts require uniqueness across all allowed values rather than one convenient example.

The hostile absence search explicitly tested these families and close-option terms: **equation; inequality; absolute value; quantitative comparison; data sufficiency; necessary; sufficient; counterexample**. A shortcut is usable only when its stated condition survives; otherwise return to the first-principles representation. Every worked answer must finish with an independent check: passage support, substitution, enumeration, units, bounds, reverse operation or option elimination.

Difficulty and timed progression

1. Foundation: recognise the family and state the governing definition or relation.
2. Core: solve a direct item with a visible representation and one verification.
3. Advanced: combine two mechanisms, test edge cases and reject close distractors.
4. Timed: use classify -> extract -> represent -> execute -> verify -> decide.
5. Remediation: log the error as concept, application, calculation, reading, passage, time or guess;

return to the owning subtopic before retesting.

BASIC MCQS / REMEDIATION

Diagnostic and core set

Q1. Practice variant 1: Solve $3x + 2 = 11$.

A. 3 B. 4 C. 2 D. 5

Correct answer: A. Subtract the constant, divide by 3, then substitute back.

Q2. Practice variant 1: Two numbers have sum 20 and difference 4. What is the larger number?

A. 13 B. 12 C. 11 D. 14

Correct answer: B. Adding the equations gives twice the larger number.

Q3. Practice variant 1: If $-2x > 8$, which statement is correct?

A. $x > -4$ B. $x < 4$ C. $x < -4$ D. $x > 4$

Correct answer: C. Dividing an inequality by a negative reverses its sign.

Q4. Practice variant 1: How many real solutions does $|x| = 4$ have?

A. 0 B. 1 C. 4 D. 2

Correct answer: D. For a positive a , $|x|=a$ gives $x=a$ and $x=-a$.

Q5. Practice variant 1: Compare Quantity I: 3^2 and Quantity II: 8.

A. Quantity I is greater B. Quantity II is greater C. They are equal D. Cannot be determined

Correct answer: A. Their difference is exactly 1.

Q6. Practice variant 1: What is x ? Statement 1: $x + 2 = 12$. Statement 2: x is positive.

A. Statement 2 alone is sufficient B. Statement 1 alone is sufficient C. Both together are required D. Even both are insufficient

Correct answer: B. Statement 1 fixes x uniquely; positivity alone does not.

Q7. Practice variant 1: What is x ? Statement 1: $x^2 = 9$. Statement 2: $x > 0$.

A. Statement 1 alone is sufficient B. Statement 2 alone is sufficient C. Both statements together are sufficient D. Even both are insufficient

Correct answer: C. Statement 1 leaves plus/minus values; Statement 2 selects the positive one.

Q8. Practice variant 1: For $0 \leq y \leq 5$, what is the maximum of $y(10-y)$?

A. 26 B. 24 C. 27 D. 25

Correct answer: D. Complete the square: $y(2n-y)=n^2-(y-n)^2$, maximised at $y=n$.

Q9. Practice variant 2: Solve $3x + 3 = 15$.

A. 4 B. 5 C. 3 D. 6

Correct answer: A. Subtract the constant, divide by 3, then substitute back.

Q10. Practice variant 2: Two numbers have sum 22 and difference 4. What is the larger number?

A. 14 B. 13 C. 12 D. 15

Correct answer: B. Adding the equations gives twice the larger number.

Q11. Practice variant 2: If $-2x > 8$, which statement is correct?

A. $x > -4$ B. $x < 4$ C. $x < -4$ D. $x > 4$

Correct answer: C. Dividing an inequality by a negative reverses its sign.

Q12. Practice variant 2: How many real solutions does $|x| = 5$ have?

A. 0 B. 1 C. 4 D. 2

Correct answer: D. For a positive a , $|x|=a$ gives $x=a$ and $x=-a$.

Q13. Practice variant 2: Compare Quantity I: 4^2 and Quantity II: 15.

A. Quantity I is greater B. Quantity II is greater C. They are equal D. Cannot be determined

Correct answer: A. Their difference is exactly 1.

Q14. Practice variant 2: What is x ? Statement 1: $x + 2 = 13$. Statement 2: x is positive.

A. Statement 2 alone is sufficient B. Statement 1 alone is sufficient C. Both together are required D. Even both are insufficient

Correct answer: B. Statement 1 fixes x uniquely; positivity alone does not.

Q15. Practice variant 2: What is x ? Statement 1: $x^2 = 16$. Statement 2: $x > 0$.

A. Statement 1 alone is sufficient B. Statement 2 alone is sufficient C. Both statements together are sufficient D. Even both are insufficient

Correct answer: C. Statement 1 leaves plus/minus values; Statement 2 selects the positive one.

Q16. Practice variant 2: For $0 \leq y \leq 6$, what is the maximum of $y(12-y)$?

A. 37 B. 35 C. 38 D. 36

Correct answer: D. Complete the square: $y(12-y)=n^2-(y-n)^2$, maximised at $y=n$.

Remediation protocol

1. Recompute without options.
2. Name the failed rule or passage phrase.
3. Reject every distractor for a specific reason.
4. Retry with altered numbers, wording or constraints.
5. Advance only after two consecutive correct answers under the time ceiling.

PYQS AND ANSWER PRACTICE

Verified 2024-2026 Set-A demand and key ledger

Year	Q	Neutral verified demand	Key status	Solution architecture
2024	35	Reversed-digit equation	D (supplied)	Use place value, cyclicity or a constrained enumeration.
2024	50	Linear-cost validation	C (supplied)	Classify the dominant mechanism, represent the givens, solve minimally and verify independently.
2024	59	Data sufficiency: integers	C (supplied)	Test each statement alone, then together, seeking uniqueness.

Year	Q	Neutral verified demand	Key status	Solution architecture
2024	60	Data sufficiency: transfer rate	C (supplied)	Test each statement alone, then together, seeking uniqueness.
2024	64	Data sufficiency: integer pair	C (supplied)	Test each statement alone, then together, seeking uniqueness.
2024	65	Data sufficiency: shares	C (supplied)	Test each statement alone, then together, seeking uniqueness.
2024	66	Data sufficiency: class size	D (supplied)	Test each statement alone, then together, seeking uniqueness.
2024	67	Data sufficiency: prime triple	A (supplied)	Test each statement alone, then together, seeking uniqueness.
2024	68	Data sufficiency: integrality	D (supplied)	Test each statement alone, then together, seeking uniqueness.
2024	69	Data sufficiency: article prices	C (supplied)	Test each statement alone, then together, seeking uniqueness.
2024	70	Data sufficiency: score comparison	D (supplied)	Test each statement alone, then together, seeking uniqueness.
2024	73	Data sufficiency: reversed ages	A (supplied)	Test each statement alone, then together, seeking uniqueness.
2024	76	Coded inequalities	D (supplied)	Translate signs and test the full allowed range.
2025	19	Chained-inequality validation	D (supplied)	Translate signs and test the full allowed range.
2025	26	Bounded-expression ratio	D (supplied)	Classify the dominant mechanism, represent the givens, solve minimally and verify independently.
2025	68	Data sufficiency: factor count	D (supplied)	Prime-factorise, count exponents and verify divisibility.
2025	69	Data sufficiency: digit product	D (supplied)	Use place value, cyclicity or a constrained enumeration.
2025	70	Data sufficiency: family relation	D (supplied)	Test each statement alone, then together, seeking uniqueness.
2025	73	Data sufficiency: match outcome	D (supplied)	Test each statement alone, then together, seeking uniqueness.
2025	74	Data sufficiency: positivity	B (supplied)	Test each statement alone, then together, seeking uniqueness.
2025	75	Quantitative comparison: ratio	D (supplied)	Classify the dominant mechanism, represent the givens, solve minimally and verify independently.
2025	76	Quantitative comparison: product	C (supplied)	Classify the dominant mechanism, represent the givens, solve minimally and verify independently.
2025	77	Quantitative comparison: set	C (supplied)	Classify the dominant mechanism, represent the givens, solve minimally and verify independently.
2025	79	Unit-interval inequalities	A (supplied)	Translate signs and test the full allowed range.
2026	1	Data sufficiency: coin denomination	A (provisional)	Test each statement alone, then together, seeking uniqueness.
2026	2	Data sufficiency: compare reals	D (provisional)	Test each statement alone, then together, seeking uniqueness.
2026	3	Data sufficiency: parity	C (provisional)	Test each statement alone, then together, seeking uniqueness.
2026	4	Data sufficiency: set-membership logic	D (provisional)	Test each statement alone, then together, seeking uniqueness.

Year	Q	Neutral verified demand	Key status	Solution architecture
2026	5	Data sufficiency: prime determination	A (provisional)	Test each statement alone, then together, seeking uniqueness.
2026	8	Weight equations comparison	B (provisional)	Classify the dominant mechanism, represent the givens, solve minimally and verify independently.
2026	21	Inequality-range statements	B (provisional)	Translate signs and test the full allowed range.
2026	38	Linear marking-scheme equations	B (provisional)	Classify the dominant mechanism, represent the givens, solve minimally and verify independently.
2026	80	Rate-constraint optimisation	D (provisional)	Use the total crossing length.

The table is a non-verbatim routing audit. It records the locally checked Set-A key letter and a valid solving route, but does not pretend that UPSC publishes model solutions. The separate workbook supplies original solved equivalents for every mechanism.

Timed mixed transfer

Q33. Practice variant 5: Solve $3x + 6 = 27$.

A. 7 B. 8 C. 6 D. 9

Correct answer: A. Subtract the constant, divide by 3, then substitute back.

Q34. Practice variant 5: Two numbers have sum 28 and difference 4. What is the larger number?

A. 17 B. 16 C. 15 D. 18

Correct answer: B. Adding the equations gives twice the larger number.

Q35. Practice variant 5: If $-2x > 8$, which statement is correct?

A. $x > -4$ B. $x < 4$ C. $x < -4$ D. $x > 4$

Correct answer: C. Dividing an inequality by a negative reverses its sign.

Q36. Practice variant 5: How many real solutions does $|x| = 8$ have?

A. 0 B. 1 C. 4 D. 2

Correct answer: D. For a positive a , $|x|=a$ gives $x=a$ and $x=-a$.

Q37. Practice variant 5: Compare Quantity I: 7^2 and Quantity II: 48.

A. Quantity I is greater B. Quantity II is greater C. They are equal D. Cannot be determined

Correct answer: A. Their difference is exactly 1.

Q38. Practice variant 5: What is x ? Statement 1: $x + 2 = 16$. Statement 2: x is positive.

A. Statement 2 alone is sufficient B. Statement 1 alone is sufficient C. Both together are required D. Even both are insufficient

Correct answer: B. Statement 1 fixes x uniquely; positivity alone does not.

Q39. Practice variant 5: What is x? Statement 1: $x^2 = 49$. Statement 2: $x > 0$.

A. Statement 1 alone is sufficient B. Statement 2 alone is sufficient C. Both statements together are sufficient D. Even both are insufficient

Correct answer: C. Statement 1 leaves plus/minus values; Statement 2 selects the positive one.

Q40. Practice variant 5: For $0 \leq y \leq 9$, what is the maximum of $y(18-y)$?

A. 82 B. 80 C. 83 D. 81

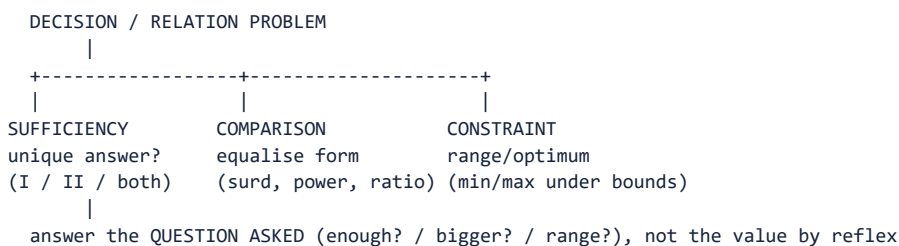
Correct answer: D. Complete the square: $y(2n-y) = n^2 - (y-n)^2$, maximised at $y=n$.

OPTIONAL ADVANCED DEPTH - NOT REQUIRED FOR A CORE ANSWER

Algebra, Inequalities and Data Sufficiency - ADVANCED

Subject: CSAT | **Tier:** Advanced Drill | **Family:** Analytical ability / decision-making. **Core skill:** unique-root traps, dependent-statement data sufficiency, quantitative comparison of surds/powers, and inequality ranges under constraints. **Grounded in:** audited UPSC CSAT PYQ sets 2024-2026 (Set A). Drills are **computationally verified**. **FACT:** = rule/fact | **ANALYSIS:** = guidance | $\square$ = original example. *Companion: Foundation.*

1. Architecture



Analytical claim: the advanced trap is **false uniqueness** - a statement that looks sufficient but allows two values (e.g., $n^2 = 49 \rightarrow n = \pm 7$), or two statements that are secretly the same.

2. Advanced tools with conditions

Tool	Statement	Condition
FACT: Unique-root test	Over the reals, $x^2 = k$ gives two roots for $k > 0$, one for $k = 0$, and none for $k < 0$; $x^3 = k$ gives one.	A squared/absolute constraint alone is often insufficient ; a cubic or odd-power one usually is not.
FACT: Dependent statements	If II is a multiple/rearrangement of I - or is logically implied by I - it adds nothing.	Two unknowns need two independent relations. Nesting can be hidden behind different-looking wording.
FACT: Necessary vs sufficient	In $A \rightarrow B$, A is sufficient for B, B is necessary for A.	Two individually necessary statements can be jointly sufficient - that is exactly the (c) verdict.
FACT: Surd comparison	Compare $\sqrt{a} + \sqrt{b}$ vs $\sqrt{c}$ by squaring (both sides positive).	Squaring preserves order only for non-negative quantities.
FACT: AM-GM / bounded optimum	For non-negative real parts with fixed sum, the product is maximised at equal parts.	AM-GM itself needs non-negative terms; for integer parts choose the nearest permitted integers, and respect any positive/distinct bounds.
FACT: Interval arithmetic	For $a < x < b$, $c < y < d$: $a+c < x+y < b+d$.	For a product on a rectangular interval, compare all four endpoint products; retain open/closed endpoints. Division additionally requires the divisor interval to exclude zero.

3. Harder methods (worked)

□ Method 1 - false uniqueness in DS

Question: What is the integer n ? **I:** $n^2 = 49$. **II:** $n^3 = 343$. I alone $\rightarrow n = 7$ or $n = -7$ (**not unique**). II alone $\rightarrow n = 7$ only (unique). So one statement (II) alone answers it but the other does not. **Answer: (a).** (Verified: $(-7)^2 = 49$ but $(-7)^3 = -343$.)

□ Method 2 - dependent statements

Question: What is $x + y$? **I:** $x + 2y = 10$. **II:** $2x + 4y = 20$. II is exactly $2 \times$ I, so it adds no information. Two unknowns, one independent equation $\rightarrow x + y$ is not determined. **Answer: (d).** (Verified: e.g. $(10,0)$ and $(8,1)$ both satisfy, giving $x+y = 10$ and 9 .)

□ Method 3 - surd comparison

Which is larger, $\sqrt{50} + \sqrt{72}$ or $\sqrt{200}$? $\sqrt{50} + \sqrt{72} \approx 7.07 + 8.49 = 15.56$; $\sqrt{200} \approx 14.14$. So $\sqrt{50} + \sqrt{72}$ is larger. (Verified.) (Note $\sqrt{50} + \sqrt{72} = 5\sqrt{2} + 6\sqrt{2} = 11\sqrt{2} \approx 15.56$, while $\sqrt{200} = 10\sqrt{2}$.)

□ Method 4 - verdict first, letter second (the escalation this paper punishes)

ANALYSIS: Advanced DS is a **two-stage** decision. Stage 1 produces a **verdict**; stage 2 maps it to a **printed letter**. The audited papers do not use one fixed mapping (see basic/05, Section 2), so never fuse the stages.

STAGE 1 - VERDICT (pure logic, no letters)
 Q alone pins it? -----> SELF-SUFFICIENT
 else I alone pins it? and II alone pins it?
 both -> EITHER ALONE
 one -> EXACTLY ONE ALONE
 none -> do they pin it TOGETHER?
 yes -> BOTH NEEDED
 no -> NEITHER SUFFICES
 STAGE 2 - MAP to whatever (a)/(b)/(c)/(d) actually say on this paper

Verdict from Stage 1	Maps to, in the audited papers
Exactly one alone	(a) in all three years.
Either alone	(b) in all three years.
Both needed	(c) in all three years.
Neither suffices	(d) where (d) reads "cannot be answered ...".
Self-sufficient question	(d) only where (d) reads "can be answered even without ..."; otherwise no option fits and you must re-read the stem for what it really asks.

- **MEMORY: Escalation rule:** if your verdict is "self-sufficient" but (d) says "cannot be answered", you have almost certainly misread the question - go back to the stem, not to the statements.

□ Method 5 - optimum and interval endpoints

Integer optimum. Positive integers x, y have $x+y=11$. Their product is largest at the nearest integers to $11/2$: $x, y = 5, 6$, so the maximum is **30**, not $11^2/4 = 30.25$ (which is the real, not integer, bound). AM-GM applies because the parts are positive.

Endpoint-aware interval. If $-2 < x \leq 3$ and $1 \leq y < 4$, the four limiting corner products are $-2, -8, 3, 12$. Thus $-8 < xy < 12$: neither extreme is reached because the responsible endpoint ($x=-2$ for the minimum and $y=4$ for the maximum) is excluded. Do not silently convert open bounds into closed ones.

4. Time-saving techniques (safe conditions)

- ANALYSIS: **Stop at uniqueness** in DS; never fully solve. *Safe always - it is the whole skill.*
- ANALYSIS: **Spot dependence** by checking whether one statement is a scalar multiple of the other. **Safe always.**
- ANALYSIS: **Factor out common surds** ($11\sqrt{2}$ vs $10\sqrt{2}$) to compare instantly. **Safe when a common radical exists.**
- ANALYSIS: **Equal-parts optimum** for a fixed-sum product. **Use AM-GM only for non-negative parts; for integers use the nearest permitted integers to the mean, then check any bounds/distinctness rule.**

5. Boundary cases

- ANALYSIS: A statement with a **square or absolute value** frequently permits **two** values - treat it as insufficient until proven unique.
- ANALYSIS: Two equations can be **inconsistent** (no solution) as well as dependent (many solutions) - both block a unique answer.
- ANALYSIS: Interval products need all four corner products even when neither interval straddles zero; then preserve whether each extremal endpoint is open or closed. Interval division is unsafe if the divisor interval contains zero.

6. Advanced traps

- WRONG: Calling $n^2 = 49$ sufficient. -> It allows $n = \pm 7$. Same for $|n - 3| = 5$.
 - WRONG: Adding a scaled duplicate equation as new information. -> It is dependent.
 - WRONG: Missing **logical** (not algebraic) dependence - II may be implied by I without looking like it.
- > Ask "does I already force II?".
- WRONG: Treating a **necessary** condition as sufficient. -> "Divisible by 3" does not give "divisible by 6"; the pair does.
 - WRONG: Multiplying corresponding interval endpoints. -> Take all four corners and preserve endpoint inclusion; never divide through an interval containing zero.
 - WRONG: Comparing surds by rough decimals when they are close. -> Square or factor the common radical.
 - WRONG: Reflexively computing a value in DS. -> Decide sufficiency and move on.
 - WRONG: Choosing the DS letter before reading what that letter says on **this** paper. -> Verdict first, letter second (Method 4).

7. Error analysis

Recurring miss	Master-Framework code	Fix
False uniqueness	A	For squares/absolute values, ask "could it be negative too?".
Missed dependence	C	Check both $II = k \times I$ and $I \Rightarrow II$.
Necessary treated as sufficient	C	Write the arrow, then read it in the right direction.
Over-solving DS	T	Stop the instant uniqueness is clear.
Interval-product sign errors	X	Tabulate the four corner products.
Surd comparison slips	X	Factor the common radical before deciding.
Right verdict, wrong letter	R	Re-read the printed option set before marking.

8. Advanced drill (with full solutions)

ANALYSIS: These items are **new** - they do not repeat Methods 1-4 above or the Foundation file. Give each a verdict **before** looking for a letter.

1. **DS.** Question: Is the real number x positive? I: $x^3 > 0$. II: $x^2 > 0$.
2. **DS.** Question: What is the two-digit number N ? I: N is a multiple of 9 and $N < 30$.

II: the digits of N sum to 9.

3. **DS.** Question: What is the integer n ? I: $|n - 3| = 5$. II: $n > 0$.
4. **DS.** Question: Is the positive integer n divisible by 6? I: n is divisible by 3.

II: n is even.

5. Given $-2 < x < 3$ and $1 < y < 4$, state the range of xy .
6. Which is larger, 5^{100} or 3^{150} ?
7. Positive integers a and b satisfy $a+b=17$. What is their maximum product?
8. Given $-1 \leq x < 2$ and $2 \leq y \leq 5$, state the range of xy .

Solutions.

1. **(a).** I: a cube keeps the sign of its base, so $x^3 > 0$ forces $x > 0$ - **sufficient**.
II: $x^2 > 0$ only rules out $x = 0$; $x = -4$ satisfies it - **insufficient**. Exactly one alone works. (Verified: $(-4)^3 = -64 < 0$ but $(-4)^2 = 16 > 0$.)
2. **"Neither suffices" -> the "cannot be answered" verdict.** I alone -> $\{18, 27\}$. II alone -> $\{18, 27, 36, 45, 54, 63, 72, 81, 90\}$. **Together** -> still $\{18, 27\}$, because II adds nothing that I did not already imply: every multiple of 9 has digit sum 9 here. Two survivors, so no unique answer even jointly. (Verified by enumeration.) This is the **hidden-dependence** trap: two statements can look independent and be logically nested.
3. **(c).** I alone -> $n = 8$ or $n = -2$ (**false uniqueness** from an absolute value). II alone -> any positive integer. Together -> $n = 8$ only. (Verified: $|8-3| = |-2-3| = 5$.)
4. **(c).** I alone: 9 is divisible by 3 but not by 6 - insufficient. II alone: 8 is even but not divisible by 6 - insufficient. Together: divisible by 2 **and** 3 forces divisibility by 6, since 2 and 3 are coprime. (Verified over 1-200.) Each statement is **necessary**; only the pair is **sufficient** - the cleanest illustration of that distinction in the whole file.
5. xy in $(-8, 12)$. The interval for x **straddles zero**, so test all four corners:
 $(-2)(1) = -2$, $(-2)(4) = -8$, $(3)(1) = 3$, $(3)(4) = 12$. Minimum -8, maximum 12. (Verified.) ANALYSIS: The naive $a \cdot c < xy < b \cdot d$ rule would have given $(-2, 12)$ and lost the true minimum - it is valid only when both intervals are positive.
6. 3^{150} . Equalise the exponent: $5^{100} = (5^2)^{50} = 25^{50}$ and $3^{150} = (3^3)^{50} = 27^{50}$, and $27 > 25$. (Verified by exact integer comparison.)
7. **72.** The nearest integers to $17/2$ are 8 and 9, so $8 \times 9 = 72$. The real equal-parts bound is 72.25, but it is unavailable to integers. (Verified.)
8. $[-5, 10)$. The corner products are -2, -5, 4, 10; -5 is attained at $x=-1, y=5$, whereas 10 would require excluded $x=2$. (Verified.)

9. Timed transfer and retry gate

Attempt the eight drills in **12 minutes**, recording each miss as C, A, X, R, or T. A DS answer is not complete until its printed option wording is checked. Retry the failed form after a 20-minute gap. Advance only at **7/8 or better**, with no false-uniqueness, denominator, or endpoint error; otherwise redo the corresponding method and take a fresh set.

10. PYQ-pattern notes (2024-2026, Set A)

- ANALYSIS: Read **Topic 05 and Topic 06 together**: the two-statement **data-sufficiency** format is a major, recurring block, and its content often overlaps with reasoning/coding (see the trend-table caveat in Master Framework).
- ANALYSIS: **The DS block is not printed the same way each year**. 2024 and 2025 attach the directions and the four options to **every item**; 2026 opens with a **single directions block covering five items**. The number of DS-format items in the audited Set-A papers is small but they arrive together, so a wrong reading of the option set costs several marks at once.
- ANALYSIS: **The fourth option changed meaning inside the 2025 paper itself** - some items offer "cannot be answered", others offer "can be answered even **without** using any of the Statements". Treat the option set as **data to be read**, not a template to be recalled.
- ANALYSIS: Recurring shapes: **data sufficiency** (unique-answer decisions), **quantitative comparison** of powers/surds, **inequality/parity** reasoning, and short **equation word problems**.
- ANALYSIS: The paper repeatedly punishes **false uniqueness** and **over-solving** - the two habits this file drills hardest.

11. Study links

- FACT: Foundation companion.
- FACT: Logical Reasoning, Coding, Counting and DI - DS content often overlaps reasoning.
- FACT: Number Systems and Number Sense - parity/factor facts settle many DS items.

Historical PYQ Integration (2018-2023)

Status: Question-level PYQ demand is integrated into this owner. **Provenance:** Audited local official-paper routing ledgers: _PYQ-ROUTING-CSAT-2018-2023.md. **Answer-key rule:** The official 2018-2023 Prelims/CSAT keys are not held locally; no option or answer has been inferred.

- **Years represented:** 2018, 2019, 2020, 2021, 2022, 2023
- **Paper(s):** CSAT
- **Routed question demands:** 41

Year	Paper	Q	PYQ demand (neutral rendering)	Directive / format	Source status	Owner requirement
2018	CSAT	2	Series equation solving	Objective question; official key unavailable locally	Stem verified against official scan; OCR artifact resolved; key unavailable locally	Practise this exact skill form under timed elimination; no answer is inferred here.
2018	CSAT	23	Range inequality deduction	Objective question; official key unavailable locally	Routed; key unavailable locally	Practise this exact skill form under timed elimination; no answer is inferred here.
2018	CSAT	24	Expression magnitude comparison	Objective question; official key unavailable locally	Routed; key unavailable locally	Practise this exact skill form under timed elimination; no answer is inferred here.

Year	Paper	Q	PYQ demand (neutral rendering)	Directive / format	Source status	Owner requirement
2018	CSAT	34	Digit-constraint algebra	Objective question; official key unavailable locally	Routed; key unavailable locally	Practise this exact skill form under timed elimination; no answer is inferred here.
2018	CSAT	37	Inequality chain deduction	Objective question; official key unavailable locally	Routed; key unavailable locally	Practise this exact skill form under timed elimination; no answer is inferred here.
2018	CSAT	38	Algebraic revenue expression	Objective question; official key unavailable locally	Stem verified against official scan; OCR artifact resolved; key unavailable locally	Practise this exact skill form under timed elimination; no answer is inferred here.
2018	CSAT	77	Equation implication deduction	Objective question; official key unavailable locally	Routed; key unavailable locally	Practise this exact skill form under timed elimination; no answer is inferred here.
2019	CSAT	18	Age ratio equations	Objective question; official key unavailable locally	Routed; key unavailable locally	Practise this exact skill form under timed elimination; no answer is inferred here.
2019	CSAT	37	Bounded inequality range	Objective question; official key unavailable locally	Routed; key unavailable locally	Practise this exact skill form under timed elimination; no answer is inferred here.
2019	CSAT	38	Age variable algebra	Objective question; official key unavailable locally	Routed; key unavailable locally	Practise this exact skill form under timed elimination; no answer is inferred here.
2019	CSAT	50	Fraction comparison property	Objective question; official key unavailable locally	Routed; key unavailable locally	Practise this exact skill form under timed elimination; no answer is inferred here.
2019	CSAT	58	Two-statement sufficiency	Objective question; official key unavailable locally	Routed; key unavailable locally	Practise this exact skill form under timed elimination; no answer is inferred here.
2019	CSAT	69	Piece-length ratio algebra	Objective question; official key unavailable locally	Routed; key unavailable locally	Practise this exact skill form under timed elimination; no answer is inferred here.
2020	CSAT	9	Equal-expression ordering	Objective question; official key unavailable locally	Routed; key unavailable locally	Practise this exact skill form under timed elimination; no answer is inferred here.
2020	CSAT	16	Digit-constraint sum puzzle	Objective question; official key unavailable locally	Routed; key unavailable locally	Practise this exact skill form under timed elimination; no answer is inferred here.
2020	CSAT	17	Two-statement sufficiency	Objective question; official key unavailable locally	Routed; key unavailable locally	Practise this exact skill form under timed elimination; no answer is inferred here.
2020	CSAT	28	Two-statement sufficiency	Objective question; official key unavailable locally	Routed; key unavailable locally	Practise this exact skill form under timed elimination; no answer is inferred here.

Year	Paper	Q	PYQ demand (neutral rendering)	Directive / format	Source status	Owner requirement
2020	CSAT	29	Two-statement sufficiency	Objective question; official key unavailable locally	Routed; key unavailable locally	Practise this exact skill form under timed elimination; no answer is inferred here.
2020	CSAT	30	Two-statement sufficiency	Objective question; official key unavailable locally	Routed; key unavailable locally	Practise this exact skill form under timed elimination; no answer is inferred here.
2020	CSAT	79	Vessel weight linear equation	Objective question; official key unavailable locally	Routed; key unavailable locally	Practise this exact skill form under timed elimination; no answer is inferred here.
2021	CSAT	8	Exponential model identification	Objective question; official key unavailable locally	Routed; key unavailable locally	Practise this exact skill form under timed elimination; no answer is inferred here.
2021	CSAT	38	Two-statement sufficiency	Objective question; official key unavailable locally	Routed; key unavailable locally	Practise this exact skill form under timed elimination; no answer is inferred here.
2021	CSAT	39	Two-statement sufficiency	Objective question; official key unavailable locally	Routed; key unavailable locally	Practise this exact skill form under timed elimination; no answer is inferred here.
2021	CSAT	47	Two-statement sufficiency	Objective question; official key unavailable locally	Routed; key unavailable locally	Practise this exact skill form under timed elimination; no answer is inferred here.
2021	CSAT	58	Two-variable score algebra	Objective question; official key unavailable locally	Routed; key unavailable locally	Practise this exact skill form under timed elimination; no answer is inferred here.
2021	CSAT	77	Age-relation algebra	Objective question; official key unavailable locally	Routed; key unavailable locally	Practise this exact skill form under timed elimination; no answer is inferred here.
2022	CSAT	19	Two-statement sufficiency	Objective question; official key unavailable locally	Routed; key unavailable locally	Practise this exact skill form under timed elimination; no answer is inferred here.
2022	CSAT	25	Two-statement sufficiency	Objective question; official key unavailable locally	Routed; key unavailable locally	Practise this exact skill form under timed elimination; no answer is inferred here.
2022	CSAT	28	Reverse chain equation	Objective question; official key unavailable locally	Routed; key unavailable locally	Practise this exact skill form under timed elimination; no answer is inferred here.
2022	CSAT	37	Two-statement sufficiency	Objective question; official key unavailable locally	Routed; key unavailable locally	Practise this exact skill form under timed elimination; no answer is inferred here.
2022	CSAT	48	Two-statement sufficiency	Objective question; official key unavailable locally	Routed; key unavailable locally	Practise this exact skill form under timed elimination; no answer is inferred here.

Year	Paper	Q	PYQ demand (neutral rendering)	Directive / format	Source status	Owner requirement
2022	CSAT	49	Two-statement sufficiency	Objective question; official key unavailable locally	Routed; key unavailable locally	Practise this exact skill form under timed elimination; no answer is inferred here.
2022	CSAT	50	Two-statement sufficiency	Objective question; official key unavailable locally	Routed; key unavailable locally	Practise this exact skill form under timed elimination; no answer is inferred here.
2022	CSAT	60	Percentage inequality comparison	Objective question; official key unavailable locally	Routed; key unavailable locally	Practise this exact skill form under timed elimination; no answer is inferred here.
2022	CSAT	78	Consecutive integer equation	Objective question; official key unavailable locally	Routed; key unavailable locally	Practise this exact skill form under timed elimination; no answer is inferred here.
2023	CSAT	26	Two-group sum product maximum	Objective question; official key unavailable locally	Routed; key unavailable locally	Practise this exact skill form under timed elimination; no answer is inferred here.
2023	CSAT	56	Two-statement inequality sufficiency	Objective question; official key unavailable locally	Routed; key unavailable locally	Practise this exact skill form under timed elimination; no answer is inferred here.
2023	CSAT	57	Two-statement integer sufficiency	Objective question; official key unavailable locally	Routed; key unavailable locally	Practise this exact skill form under timed elimination; no answer is inferred here.
2023	CSAT	58	Two-statement set-size sufficiency	Objective question; official key unavailable locally	Routed; key unavailable locally	Practise this exact skill form under timed elimination; no answer is inferred here.
2023	CSAT	59	Two-statement digit-number sufficiency	Objective question; official key unavailable locally	Stem verified against official scan; OCR artifact resolved; key unavailable locally	Practise this exact skill form under timed elimination; no answer is inferred here.
2023	CSAT	60	Two-statement age-sequence sufficiency	Objective question; official key unavailable locally	Stem verified against official scan; OCR artifact resolved; key unavailable locally	Practise this exact skill form under timed elimination; no answer is inferred here.

What this owner must now support

- Series equation solving
- Range inequality deduction
- Expression magnitude comparison
- Digit-constraint algebra
- Inequality chain deduction
- Algebraic revenue expression
- Equation implication deduction
- Age ratio equations
- Bounded inequality range
- Age variable algebra
- Fraction comparison property

- Two-statement sufficiency
- Piece-length ratio algebra
- Equal-expression ordering
- Digit-constraint sum puzzle
- Vessel weight linear equation
- Exponential model identification
- Two-variable score algebra
- Age-relation algebra
- Reverse chain equation
- Percentage inequality comparison
- Consecutive integer equation
- Two-group sum product maximum
- Two-statement inequality sufficiency
- Two-statement integer sufficiency
- Two-statement set-size sufficiency
- Two-statement digit-number sufficiency
- Two-statement age-sequence sufficiency

The table integrates the examinable demand and paper metadata. It does not turn an unkeyed objective question into a solved answer, and it does not claim that lexical presence alone proves full conceptual sufficiency.

CONSOLIDATED REGISTER NOTES

Complete revision spine

- **Expression discipline:** Track signs, brackets, domains and denominators.
- **Linear equations:** Preserve equality through reversible operations.
- **Simultaneous equations:** Use elimination, substitution or option testing.
- **Inequalities:** Reverse the sign only when multiplying or dividing by a negative.
- **Absolute value and surds:** Split valid cases and respect non-negative roots.
- **Word translation:** Define variables and constraints before manipulating.
- **Quantitative comparison:** Compare ranges, not one sample.
- **DS format:** Read the printed verdict options before solving.
- **Necessary and sufficient:** Test each statement alone, then together.
- **Counterexample search:** One second valid value disproves uniqueness.
- **Bounds and optimisation:** Check endpoints, integrality and attainability.
- **Verification:** Substitute, test domains and separate answer value from sufficiency.

Ownership and close-option firewall

- **Own here:** Owns algebraic expressions, equations, inequalities, quantitative comparison and the two-statement data-sufficiency decision format.
- **Do not duplicate:** Arithmetic computation remains with Topic 03 and rate contexts with Topic 04 when no sufficiency judgement is tested; arrangements and coding belong to Topic 06.
- **Verification:** Equations are plug-checked, inequality sign changes are explicit, and sufficiency verdicts require uniqueness across all allowed values rather than one convenient example.

Timed answer route

CLASSIFY -> EXTRACT -> REPRESENT -> EXECUTE -> VERIFY -> DECIDE

- Use estimation or option elimination only after preserving the governing condition.
- A blank costs zero; a rushed unsupported answer also consumes time and may attract negative marks.
- For every error, record concept/application/calculation/reading/passage/time/guess, repair the owner, and retry a new item rather than memorising the old option.